

WaveCast

Surf & Wave Forecast Web Application

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1. Project Overview

WaveCast is a full-stack surf and wave forecasting web application inspired by Surfline. The platform provides real-time and 7-day wave forecasts for any coastal location worldwide, combining multiple live data APIs with a persistent backend, user authentication, and intelligent surf condition scoring. The project is designed to demonstrate production-grade web development skills, including RESTful API integration, relational database design, caching strategies, and interactive data visualization.

2. Motivation

Surfers and ocean sports enthusiasts rely on expensive subscription services (e.g., Surfline) to access wave data. WaveCast aims to provide a free, open-source alternative with a clean modern interface, powered entirely by free public APIs. Beyond the domain, the project serves as an end-to-end demonstration of a scalable web architecture with real data flows.

3. Core Features

Feature	Description
Wave Forecast	7-day forecast: wave height, period, direction
Surf Rating	Algorithmic Good/Fair/Poor/Epic scoring
Interactive Map	Click anywhere on the coast - Leaflet.js
Forecast Charts	Time-series graphs via Recharts
User Auth	JWT-based registration & login
Favorites	Save/manage favorite surf spots (DB)
Alerts	Email alert when waves meet user criteria
API Caching	DB-level cache to reduce external API calls

4. Technology Stack

Layer	Technology	Purpose
Frontend	React + Vite + Tailwind CSS	UI framework & styling
Charts	Recharts	Time-series forecast graphs
Map	Leaflet.js	Interactive location picker
Backend	Node.js + Express	REST API server
ORM	Prisma	DB schema & queries
Database	PostgreSQL (Supabase)	Users, favorites, cache
Auth	JWT + bcrypt	Secure authentication
APIs	Open-Meteo Marine, OpenWeatherMap	Wave & weather data
Email	Nodemailer + SendGrid	Wave alerts
Deploy	Vercel + Railway	Frontend + backend hosting

5. System Architecture

The application follows a decoupled client-server architecture. The React frontend communicates with an Express backend via a REST API. The backend is responsible for all external API calls, authentication logic, and database access. PostgreSQL (hosted on Supabase) stores users, saved spots, alert preferences, and a forecast cache table. A caching layer prevents redundant calls to the weather APIs by storing results with a timestamp and serving cached data if it is less than 3 hours old.

Component	Responsibility
React Client	Renders UI, map, charts; sends auth tokens
Express Server	Routes, auth middleware, business logic
Prisma + Postgres	Persists users, favorites, cached forecasts
Open-Meteo API	Wave height, period, direction (free, no key)
OpenWeather API	Wind speed, temperature, UV index
SendGrid	Transactional email for wave alerts

6. Surf Rating Algorithm

A key differentiator of WaveCast is a rule-based surf quality scoring engine. Each forecast hour receives a score (0-100) computed from the following weighted parameters:

Parameter	Weight	Ideal Range
Wave Height (m)	35%	1.0 - 2.5 m
Wave Period (s)	30%	> 10 s

Parameter	Weight	Ideal Range
Wind Speed (km/h)	20%	< 15 km/h (offshore preferred)
Wave Direction	15%	Perpendicular to coastline

The score maps to a label: 0-39 = Poor, 40-59 = Fair, 60-79 = Good, 80-100 = Epic. Users can set alert thresholds (e.g., notify me when score ≥ 70 at a saved spot).

7. Development Work Plan

Phase	Tasks
1 - Setup	Monorepo init, ESLint, Prettier, Supabase project, CI skeleton
2 - API Integration	Connect Open-Meteo Marine + OpenWeather, parse responses, unit tes
3 - Map + UI	Leaflet map, location picker, basic forecast cards
4 - Charts	Recharts time-series for height, period, wind; surf rating badge
5 - Backend + DB	Express routes, Prisma schema (User, Spot, ForecastCache), migration
6 - Auth	JWT registration/login, protected routes, token refresh
7 - Favorites + Alerts	Save spots, alert preferences table, SendGrid email trigger
8 - Caching	DB-level forecast cache with TTL, cache-hit logging
9 - Deploy + Polish	Vercel + Railway deploy, README, env docs, final QA

8. Complexity Justification

The following technical elements elevate WaveCast beyond a basic API-display project:

- **Multi-source API aggregation** - merges marine and weather data into a unified response object.
- **Database caching layer** - reduces external API calls and demonstrates production-aware design.
- **JWT authentication flow** - full registration, login, token validation, and protected routes.
- **Dynamic surf rating engine** - multi-parameter weighted algorithm with user-configurable alerts.
- **Interactive map** - arbitrary coordinate selection rather than a fixed dropdown list.
- **Email notification system** - async alert pipeline triggered by a scheduled check (cron job).
- **Full deployment** - separate frontend/backend hosting with environment variable management.

9. External APIs

API	Data Provided	Cost
Open-Meteo Marine	Wave height, period, direction	Free, no key
OpenWeatherMap	Wind, temperature, UV, humidity	Free tier (1000 req/day)
SendGrid	Transactional email delivery	Free tier (100 emails/day)

10. Summary

WaveCast is a well-scoped, technically rich project that covers the full web development stack: interactive React frontend, RESTful Node.js backend, relational database with Prisma ORM, JWT authentication, multi-API data aggregation, intelligent caching, and automated email alerts. The surf rating algorithm provides a domain-specific layer of logic that goes beyond simple data display. The project will be fully deployed and publicly accessible by the submission deadline.